The Use of Polls to Enhance Formative Assessment Processes in Mathematics Classroom Discussions

Annalisa Cusi, Francesca Morselli and Cristina Sabena

1 Introduction and Background

Formative assessment (FA) or assessment for learning is generally conceived as a teaching method, where "evidence about student achievement is elicited, interpreted, and used by teachers, learners, or their peers, to make decisions about the next steps in instruction that are likely to be better, or better founded, than the decisions they would have taken in the absence of the evidence that was elicited" (Black & Wiliam, 2009, p. 7). Large-scale reviews have revealed the effectiveness of educational interventions that focus on the development of teaching using FA in classrooms compared to most other intervention approaches (Hattie, 2009). However, FA practices constitute a great challenge for teachers in the classroom and research has started investigating how technology can support them: this was one of the goals of the European project FaSMEd—Improving Progress for Lower Achievers through Formative Assessment in Science and Mathematics Education). Within FaSMEd, we focused in particular on the role that the so-called connected classroom technologies (CCT) may play in FA mathematics classroom processes at primary and lower secondary school levels (Cusi, Morselli, & Sabena, 2017).

CCT are networked systems of computers or handheld devices specifically designed to be used in a classroom for interactive teaching and learning. Previous research has underlined those affordances of CCT that make them effective tools for FA: monitoring students' progress, collecting the content of students' interaction

A. Cusi (✉)
Sapienza University of Roma, Rome, Italy
e-mail: annalisa.cusi@uniroma1.it

F. Morselli
University of Genova, Genoa, Italy

C. Sabena
University of Torino, Turin, Italy

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over long timespans and for multiple sets of classroom participants (Roschelle & Pea, 2002); providing students with immediate private feedback, supporting them with appropriate remediation and keeping them oriented on the path to deep conceptual understanding (Irving, 2006); enabling students to take a more active role in classroom discussions and encouraging them to reflect and monitor their own progress (Roschelle & Pea, 2002; Ares, 2008).

Notwithstanding the potential of these tools, many researchers have stressed that their effectiveness depends on the skill of the instructor and on his/her ability to incorporate procedures such as tracking students' progress, keeping students motivated and enhancing reflection with technologies (Irving, 2006; Kay & Le Sage, 2009). Some studies, in particular, have highlighted that CCT increase the complexity of the teacher's role with respect to 'orchestrating' the lesson (Clark-Wilson, 2010; Roschelle & Pea, 2002). Therefore, in order to bring about progress in student participation and achievement, technology must be used in conjunction with particular kinds of teaching strategies. To this respect, Beatty and Gerace (2009) developed technology-enhanced formative assessment (TEFA), a pedagogical approach for teaching science and mathematics with the aid of classroom response system (CRS). CRS[1] consist of a set of input devices for students, communicating with the software running on the instructor's computer, and enabling the instructor to pose questions to students and take a follow-up poll (Beatty & Gerace, 2009).

To help teachers implement FA, the TEFA approach (Beatty & Gerace, 2009) introduces an iterative cycle of question posing, answering, and discussing, which forms a scaffold for structuring whole-class interaction with the aid of CRS. The essential phases of the cycle are:

- pose a challenging question to the students;
- have students wrestle with the question and decide upon a response;
- use a CRS to collect responses and display a chart of the aggregated responses;
- elicit different reasons and justifications from students for the chosen responses;
- develop a student-dominated discussion of the assumptions, perceptions, ideas, and arguments involved;
- provide a summary, micro-lecture, and meta-level comments.

From the 1970s to the 2010s, many research studies on the use of CRS in educational settings have been developed, as documented in meta-reviews that highlighted benefits and challenges of CRS-integrated instruction (Fies & Marshall, 2006; Kay & Le Sage, 2009; Chien, Chang, & Chang, 2016; Hunsu, Adesope, & Bayly, 2016).

Among the benefits of CRS-integrated instruction, research indicates that CRS simultaneously provide anonymity and accountability, support collecting answers from all students in a class rather than just the few who speak up or are called upon, and enable recording data of students' individual and collective responses for subsequent analysis (Beatty & Gerace, 2009).

[1] In research literature different terms are used to refer to these devices, such as clickers, Audience Response Systems, Student Response Systems, Interactive Response Systems.

Research has also identified different factors that contribute in increasing the quantity and quality of class discussions, when CRS are used: asking subjects to generate explanations and justifications for their own answers to the questions posed by means of CRS (Chien et al., 2016); engaging students in peer discussion (Chien et al., 2016); and collecting students' responses and presenting them to the class without providing the correct answer (Kay & Le Sage, 2009).

Beatty and Gerace (2009) highlight that CRS may be exploited by the teacher with great flexibility, and list specific instructional purposes connected to their use. Among them are the uses of polls for:

- status check, that is to ask students their self-reported degree of confidence in their understanding of a topic;
- exit poll, that is to poll students to find out which concepts they want to spend more time on;
- assess prior knowledge, that is to elicit what students know or believe about a topic;
- provoke thinking, that is to ask a question to get students engaged within a new topic;
- elicit a misconception, that is to lead students to manifest a specific common misconception or belief that may hinder their learning;
- exercise a cognitive skill, that is to engage students in a specific cognitive activity;
- stimulate discussion with questions having multiple reasonable answers;
- review, that is to pose questions aimed at reminding students of a body of material already covered.

Meta-reviews also highlight specific aspects on which future research should focus. First of all, most of the meta-reviews stress the need for further research on the use of CRS in K-12 classrooms, since most of the examined studies used undergraduates or graduates as research samples (Kay & Le Sage, 2009; Chien et al., 2016; Hunsu et al., 2016). Others focus on the methodologies through which CRS are used in the classroom and, in particular, on the ways in which students are engaged in CRS-integrated instruction. Fies and Marshall (2006), for example, suggest deeply exploring the effects of group mode use of CRS. In tune with this idea, Kay and Le Sage (2009) suggest studying the influence of students' comparison with peers on developing a classroom community. Finally, we remark that another important feature highlighted is the need to explore how CRS could be used in order to promote meta-cognitive and self-regulatory learning strategies in classes (Hunsu et al., 2016).

Against this background, in our study we focused on the use of polls to enhance effective classroom discussions with FA purposes at primary and lower secondary school levels.

2 Formative Assessment with Technology: A Theoretical Framework

Wiliam and Thompson (2007) identified five key strategies for FA:

(A) Clarifying and sharing learning intentions and criteria for success;
(B) Engineering effective classroom discussions and other learning tasks that elicit evidence of student understanding;
(C) Providing feedback that moves learners forward;
(D) Activating students as instructional resources for one another;
(E) Activating students as the owners of their own learning.

These strategies may be activated by three agents: the teacher, the peers and the student themselves. Technology, indeed, may support the three agents in activating the FA strategies in different ways.

Within the FaSMEd project, we developed a three-dimensional framework for the design and implementation of technologically-enhanced formative assessment activities. The framework is represented in the chart[2] in Fig. 1 and extends Wiliam and Thompson's model (whose dimensions are the five key strategies for FA and the agents), adding to it the dimension related to the functionalities through which technology could support FA (Aldon, Cusi, Morselli, Panero, & Sabena, 2017; Cusi et al., 2017). These functionalities are:

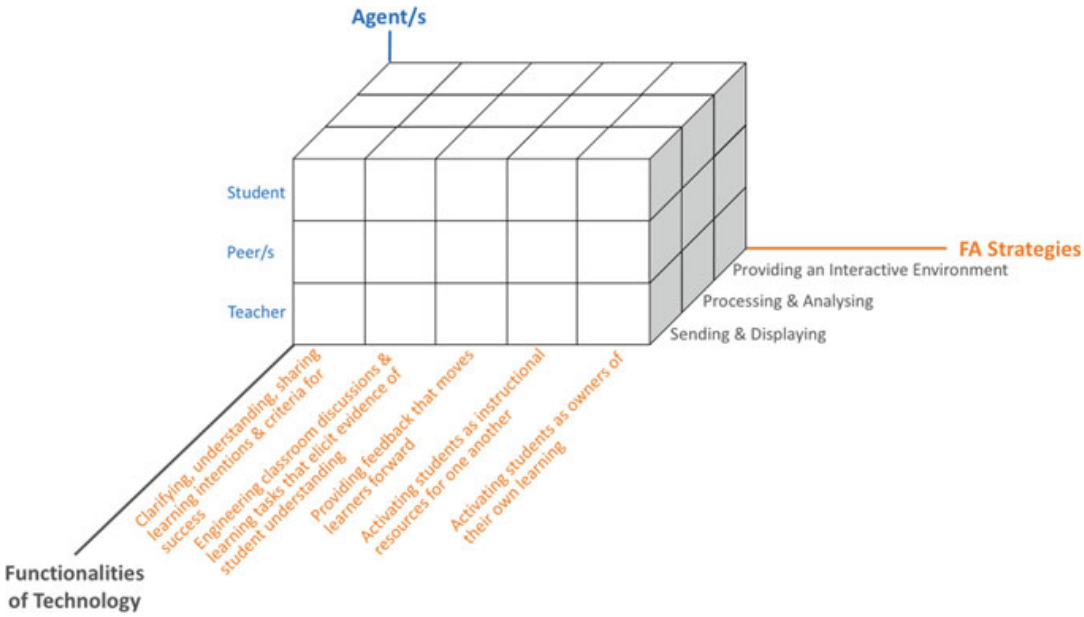


Fig. 1 Chart of the FaSMEd three-dimensional framework (the three dimensions—FA strategies, agents, functionalities of technology—are represented in the three axes of the diagram)

[2]We thank D. Wright (Newcastle University) for the digital version of the chart and Hana Ruchniewicz (University of Duisburg-Essen) for its adaptation.

(1) *Sending and displaying*, that is the ways in which technology supports the communication among the agents of FA processes (e.g., sending and receiving messages and files, displaying and sharing screens or documents to the whole class).
(2) *Processing and analysing*, that is the ways in which technology supports the processing and the analysis of the data collected during the lessons (e.g., through the sharing of the statistics of students' answers to polls or questionnaires, feedback given directly by the technology to the students when they are performing a test).
(3) *Providing an interactive environment*, that is when technology enables the creation of environments in which students can interact to work individually or in groups on a task or to explore mathematical/scientific contents (e.g., through the creation of interactive boards to be shared by teacher and students or the use of specific software that provides an environment where it is possible to dynamically explore specific mathematical problems).

A fundamental aspect on which FA is focused is feedback, defined by Hattie and Timperley (2007) as "information provided by an agent (e.g. teacher, peer, book, parent, self, experience) regarding aspects of one's performance or understanding" (p. 81).

In our design and in the data analysis we refer, in particular, to the four major levels of feedback introduced by Hattie and Timperley (ibid.):

- feedback about the task, which concerns how well a task is being accomplished or performed;
- feedback about the processing of the task, which concerns the processes underlying tasks or relating and extending tasks;
- feedback about self-regulation, which refers to the way students monitor, direct, and regulate actions toward the learning goal;
- feedback about the self as a person, which consists in positive (and sometimes negative) evaluations of and effects on the student.

3 Designing FA Activities Within a CCT Environment

The study documented in this paper is part of a wider design-based research, characterized by cycles of design, enactment, analysis and redesign, where the goal of designing learning environments is intertwined with that of developing new theories (DBRC, 2003). The research is carried out in authentic settings (classroom environments) focusing on "interactions that refine our understanding of the learning issues involved" (DBRC, 2003, p. 5).

In our design study, in tune with the FA framework, we focus on the crucial role of the interaction with peers and with an expert in students' learning. In line with Vygotskian perspectives (Vygotsky, 1978), we consider effective mathematical

discussions (Bartolini Bussi, 1998) as fundamental activities, where the teacher plays a key role in planning and promoting fruitful occasions for FA and learning.

Moreover, we also believe that FA has to focus on affective (Hannula, 2011) and metacognitive (Schoenfeld, 1992) factors. Accordingly, we designed activities aimed at supporting students in (a) making their thinking visible (Collins, Brown, & Newmann, 1989) through the sharing of their thinking processes with the teacher and classmates by means of argumentative processes, (b) developing their ongoing reflections on the learning processes and (c) discussing their emotions during the FA activity.

An important feature of the task design is a strong argumentative component: students are always required to explain their answers in a written text. Students' argumentations are then collectively analysed according to three criteria: the correctness (Do these justifications contain any mistake?), the clearness (Is every reader able to easily understand these justifications?) and the completeness (Do these justifications contain all the information necessary to draw these conclusions?) of the justifications provided by the students.

Concerning technology, we explored the use of a CCT (provided by a software called IDM-TClass), which connects the students' tablets with the teacher's laptop, allows the students to share their productions and the teacher to easily collect the students' opinions and reflections, during or at the end of an activity, by means of the creation of instant polls.

The use of IDM-TClass was integrated within a set of activities on relations and functions as well as their representations (symbolic representations, tables, graphs) adapted from different sources (the ArAl Project: www.progettoaral.it; and the Mathematics Assessment Program: http://map.mathshell.org). For each activity, we designed a sequence of worksheets (doc files), to be sent to the students' tablets or to be displayed on the interactive whiteboard (or through the data projector).

The worksheets were designed according to three main categories: (1) worksheets introducing a problem and asking one or more questions (problem worksheets); (2) worksheets aimed at providing support to students who met difficulties in facing the given tasks (helping worksheets); and (3) worksheets prompting a poll between proposed options (poll worksheets).

Concerning the poll modality, IDM-TClass software collects all the students' choices and processes them, displaying an analytical record (collection of each answer) as well as a synthetic overview (bar chart). In reference to the analytical framework, instant polls are used through the support of the "Processing and Analysing" functionality of technology. The possibility of showing the results in real time also brings to the fore the "Sending and Displaying" functionality of technology.

In principle, the software also enables the teacher to set the time given to students before completing the poll, and offers the opportunity to provide an immediate automatic correction to the student. However, our choice is not to provide an immediate automatic correction to students, so that they can instead be engaged in a subsequent classroom discussion. In tune with Beatty and Gerace's (2009) framework, in fact, we conceive the use of polls as a way of scaffolding whole-class interaction with

the aim of fostering the sharing of results and the comparison between students (FA strategy B). This is also coherent with our belief on the key role of the teacher and the importance of peer interaction.

During the design experiments, we implemented planned polls and instant polls. Planned polls were created a priori and were part of each teaching sequence. They were realized through poll worksheets, which can be used alternatively to problem worksheets. Instant polls were on the contrary created and implemented on the spot during the lesson. In design-based research, instant polls that prove fruitful in terms of FA strategies may be inserted in the repertoire of planned polls for subsequent cycles of experimentation.

4 Categories of Polls

In our design, polls are always intended as a starting point for class discussion and not for individual "revising" or "status check". After three cycles of design, implementation and analysis of the classroom activities, we classified polls according to the different foci and aims of the classroom discussions developed from them.

We identified four categories of polls, which are presented in Table 1, together with the corresponding aim and an example from our design experiments. Two examples will be presented in an extended way in the data analysis section.

Referring to the instructional purposes of polls described by Beatty and Gerace's framework (2009), polls belonging to category 1 may be related to the aims to "provoke thinking" and "exercise a cognitive skill", whereas the polls belonging to category 2 may be linked to the aims to "elicit a misconception" and "stimulate discussion with questions having multiple reasonable answers". Polls belonging to categories 3 and 4 are of different nature: even if they could be somehow related to "status check", they bring to the fore metacognitive and affective issues that are not so evident in Beatty and Gerace's list.

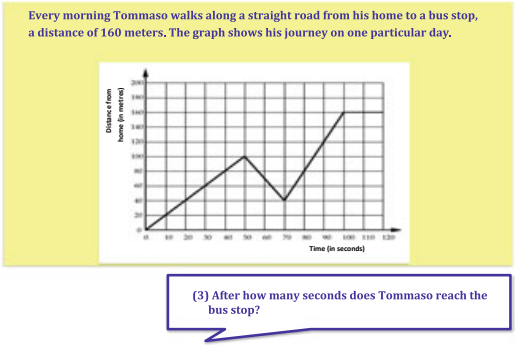
5 Research Questions and Methodology

Concerning polls, our investigation is guided by the following research questions:

- *Which FA strategies can be activated by means of technologically enhanced polls?*
- *What are the main characteristics of the FA discussions developed by means of technologically enhanced polls? i.e.,: How could they be initiated? How could they evolve?*

All the lessons were video-recorded, fields notes were taken and students' productions (doc files) were collected, building a large amount of data (about 450 h of class sessions, carried out in collaboration with 20 teachers).

Table 1 Categories of polls resulting from our design experiments

Category of poll	Corresponding aims	Example from our design experiments
(1) Polls on specific mathematical content: these polls ask to choose the correct answer to a problem or to a specific question	– To highlight students' understanding of specific topics developed during the lesson – To make students discuss the reasons linked to the choice of the correct answers (providing feedback about the task) – To promote a discussion on the solving strategies, in order to enable students to share and compare them (providing feedback about the processing of the task)	Every morning Tommaso walks a straight road from his home to a bus stop, a distance of 160 m. The graph shows his journey on one particular day  After how many seconds does Tommaso reach the bus stop? (a) After 120 s (b) After 50 + 70 + 100 + 120 s, that is after 340 s (c) After 100 s (d) After 50 s

(continued)

Table 1 (continued)

Category of poll	Corresponding aims	Example from our design experiments
(2) Polls on argumentation: these polls ask students to compare different justifications about the answers to a given problem	– To promote a discussion at a meta-mathematical level, focused on the way the answer is justified. Thanks to this kind of discussion, students are led to identify the criteria to assess the answers. Hence, students receive and provide feedback on the processing of the task and feedback about self-regulation	See Sect. 7
(3) Polls on metacognitive aspects: these polls are focused on the difficulties students meet when facing specific kind of tasks or on the best strategies to be used to face specific tasks	– To promote metacognitive reflections that could help students in identifying the available tools to face similar tasks in the future and in becoming aware of how to monitor themselves while facing this kind of task (providing *f*eedback about self-regulation)	See Sect. 8
(4) Polls on affective aspects: these polls are focused on students' feelings when facing a specific kind of task or when a particular methodology is adopted during the lessons	– To bring to the fore the affective dimension, supporting students in becoming aware of their way of posing themselves during this kind of activity (providing feedback about self-regulation)	How did you feel when your answer was displayed on the interactive whiteboard? A) Uneasy B) Happy C) Calm D) Worried

In line with design-based research, the study is carried out through close collaboration between researchers and teachers, who share the aim of improving practice, taking into account both contextual constraints and research aims.

At least one researcher was always present in the classroom as participant observer during the design experiments.

The analysis of the video-recordings of class discussion was developed according to the following methodology:

- a preliminary selection of class sessions was made on the basis of researchers' direct observations;
- the selected classroom discussion episodes were transcribed and analysed separately by the researchers, who coded the transcripts in terms of FA strategies;
- problematic codes were discussed together so that researchers could come to an agreement.

6 Data Analysis

Hereunder we present two examples taken from our experiments. Both examples refer to a task sequence on time-distance graphs adapted from the task sequence "Interpreting time-distance graphs", from the Mathematics Assessment Program (http://map.mathshell.org/materials/lessons.php). From the original source based on paper and pencil materials for grade 8, we adapted the activities and created a set of 19 worksheets to be used with students from grades 5–7 (ages 10–12).

The sequence starts with a short text about the walk of a student, Tommaso, from home to the bus stop. This text is accompanied by a time-distance graph, as illustrated in Fig. 2.

Students' interpretation of this graph is guided through questions, posed to them within problem, helping and poll worksheets. After interpretation of the time-distance graph according to the given story, the activity develops through the matching of different graphs and the corresponding stories and the construction of graphs associated to specific stories. Since this was students' first encounter with time-distance graphs, we designed an introductory activity based on the use of a motion sensor, in which students could explore in a laboratorial way the construction of the graph after a motion experience along a straight line. The students worked in two ways: first, they walked and observed the graph of their motion provided by the motion sensor; after, they were given some graphs and were asked to walk so as to obtain the same graph by means of the motion sensor. An inspiring reference for the design of this introductory activity was the research on children's use of motions detectors to make sense of Cartesian graphs of position versus time (see, for instance, Nemirovsky, Tierney, & Wright, 1998).

7 Example 1: A Classroom Discussion from a Poll on Argumentation

In the following, we present an excerpt from a grade 7 class discussion starting from a poll belonging to category 2: Polls that ask to compare different answers to a problem. The episode concerns the interpretation of the final part of the graph in Fig. 2. At first, students were asked via a problem worksheet to establish what happens during the last 20 s, and to justify their answers. During the classroom discussion, a poll worksheet was used to focus on the completeness of answers, which may be referred to as FA strategy A (Clarifying and sharing learning intentions and criteria for success). Specifically, the poll required students to identify which is the most complete among three given answers:

Some students of another class wrote these answers. Which of them is the most complete?

(A) During the last 20 s, Tommaso is not walking because we have already said that he has reached the bus stop.

(B) I think that, during the last 20 s, Tommaso is not walking because from the graph it is possible to understand that, in the period between 100 s and 120 s, he is always at the same distance from home, that is 160 m.

(C) I understood that, during the last 20 s, Tommaso is not walking because the line of the graph is horizontal.

Option B represents the most complete answer to the question about what happens to Tommaso during the last 20 s because it refers to the correct interpretation of the graph in terms of time and corresponding distance from home. Option A is the typical

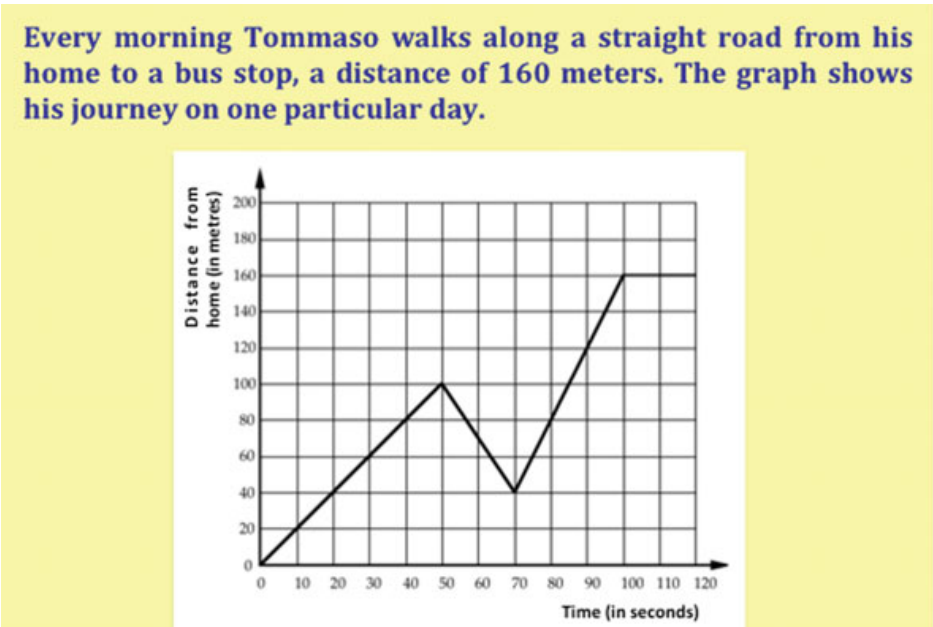


Fig. 2 The time-distance graph of Tommaso's walk

justification provided by a student who is not referring to the graph, but only to their knowledge about "the end of the story" (Tommaso reaches the bus stop). Option C could be the typical answer of those students who prefer to refer to their previous experience with the motion sensor (through which they discovered that a horizontal line represents the fact that an object is not moving), instead of trying to understand how a conclusion could be drawn through a deeper interpretation of the graph.

In order to answer to the poll, students discussed the question in pairs. After all the pairs sent their answers, the teacher displayed the distribution of answers on the IWB: 10% of the students chose option A, 50% chose option B and 40% chose option C. Starting from the display of the results, the discussion took place. The teacher exploited the poll worksheet as a way to engineer effective classroom discussions that elicit evidence of student understanding (FA strategy B). Table 2 presents selected excerpts from the discussion, analysed according to the FaSMEd framework.

Table 2 Excerpts from the class discussion and corresponding analysis

Excerpts from the class discussion	Analysis according to the FaSMEd three-dimensional framework
After brief analysis of A, justifications B and C are compared (353) Teacher: let's look at B and C. Let's hear some explanations of those who chose C, why they chose C and some motivations of those who chose B (354) Brown: we chose B because B specifies also that he (Tommaso) stayed still from 100 to 120 s, while C doesn't say this, saying that they were only 20 s; they could have been 150, 170, 180 and so on… (355) Silvia: B is the most complete (356) Teacher: B is the most complete (357) Mario: for me B is not right because, when we used the motion sensor, let's say, you understand that a person stops when the line is horizontal, and there (justification B) it doesn't say this, meaning it is not the most complete	The teacher encourages the students to discuss the reasons behind the choices of the poll. Her aim is to promote a discussion on the completeness of the two options. This is an instance of FA Strategy A, since the focus is on the requirements that a complete answer must satisfy Suggesting that answer B gives more information on the last trait, Brown activates herself as responsible for her learning (FA strategy E) and at the same time as instructional resource for her classmates (FA strategy D). Silvia, echoing Brown, affirms that B is the most complete, thus giving implicit feedback to Brown (FA strategy C). In line 357 Mario challenges the former evaluation, activating himself as owner of his own learning (FA strategy E): in his opinion, answer B is not complete because it does not refer to the experience with sensor detectors. This intervention provides a good opportunity to discuss again the role and value of the empirical experience with sensors

(continued)

Table 2 (continued)

Excerpts from the class discussion	Analysis according to the FaSMEd three-dimensional framework
(390) Lollo: but if we had not done that activity before… (391) Teacher: the activity with the motion sensor (392) Lollo: we could not have known that if you are still the line is horizontal	Lollo suggests that one cannot refer to the experience with sensors, since the answer should also be intelligible by a reader who did not have such an experience. Lollo turns himself as an instructional resource for his classmates (FA strategy D), in particular giving feedback to Mario (FA strategy C). The teacher reformulates Lollo's intervention so as to involve the other students, turning Lollo into a resource for his classmates (FA strategy D). In this way, she also activates FA Strategy C
(399) Rob: And, anyway, from the graph you can understand why the distance is always the same but the seconds, let's say, go on… (400) Teacher: ok… then, even if we had not had the experience with the motion sensor, that made you understand in an experimental way that if I stay still the line is horizontal, your classmate [Rob] says: "from the graph I can understand it anyway". Why? Rob, could you please repeat it? (401) Rob: because from the graph you can understand that when you don't move, that is to say when there is the horizontal line… (402) Teacher: what does it mean? (403) Rob: the metres remain the same but the seconds go on, let's say	Rob intervenes, stating that in the horizontal trait the distance from home is always the same. This is a shift from an explanation based on the experience with sensors to a theoretical explanation, based on the meaning of the graph. Rob provides other students with feedback to move forward (FA strategy C), turning himself into an instructional resource for his classmates (FA strategy D) The teacher reformulates Rob's intervention, giving all the students feedback that moves them forward (FA strategy C). Reformulation is also a means to activate Rob as a resource for his classmates (FA strategy D)
(413) Teacher: B explains why the line is horizontal, while C just says "the line is horizontal"; B instead explains why the line is horizontal, because the metres remain the same, even if time goes on, isn't it?	As a final intervention, the teacher rephrases the result of the discussion, pointing out what makes answer B more complete than the other options. In this way, she activates FA strategy A

8 Example 2: A Discussion from a Poll on Metacognitive Aspects

In the following, we present an excerpt from a grade 5 class discussion starting with a poll belonging to category 3: Polls on metacognitive aspects. The discussion developed from the results of an instant poll. The instant poll was proposed at the end of the task sequence on time-distance graphs, created on the spot by the teacher (T) and researcher (R) with the aim of boosting a metacognitive reflection on effective ways to tackle graph interpretation tasks. Here is the wording of the poll:

"When interpreting a graph, what is the first thing you look at?"

(a) If the graph starts from the origin
(b) If the graph goes up or down
(c) If the graph has horizontal traits
(d) How many traits compose the graph
(e) How steep is the graph
(f) What is written on the axes.

We may note that, different from the poll in example 1, this poll does not encompass only one correct answer. The subsequent discussion is aimed at making students' strategies visible when approaching a graph and comparing the efficiency of such strategies.

Most students (72%) chose F ("What is written on the axes"); 18% chose A ("If the graph starts from the origin") and 9% chose C ("If the graph has horizontal traits") (see Fig. 3 for the representation of the results that was displayed to the class).

Starting from the display of the results, the discussion took place. The teacher exploited the poll as a way to engineer effective classroom discussions that elicited evidence of student understanding (*FA strategy B*). Table 3 presents selected excerpts from the discussion, analysed according to the FaSMEd framework.

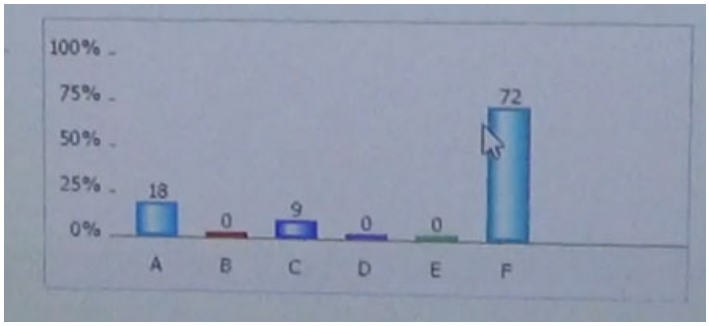


Fig. 3 Results of the instant poll, as displayed on the IWB

Table 3 Excerpts from the class discussion and corresponding analysis

Excerpts from the class discussion	Analysis according to the FaSMEd three-dimensional framework
1. Researcher: Here we have 72% that answered F 2. Teacher: the axes 3. Researcher: "What is written on the axes". Someone chose A: "If the graph starts from the origin". Someone chose C: "If there are horizontal traits". The other options were not chosen. Some of you said you changed her mind. Would you like to tell me your answer now? (speaking to Sabrina) 4. Sabrina: We chose A, but later we changed our mind. We want to choose F 5. Researcher: So, actually for you it is F?	The functionality of technology is processing and analysing, since the display of data is the starting point for a discussion. The activated FA strategy is B (engineering effective classroom discussions) Immediately after the display of the results, Sabrina and her classmate ask to change their choice: this can mean that they recognise they have answered without deep reflection. Recognising this and asking to amend the answer is an instance of FA strategy E (they make themselves responsible for their own learning)

(continued)

Table 3 (continued)

Excerpts from the class discussion	Analysis according to the FaSMEd three-dimensional framework
6. Researcher: We could start from F. Why do you think the first thing to look at is what is written on the axes? 7. Some students raise their hands 8. Elsa: Because, if you look at what is written on the axes, you can already understand the graph… and you can get some information 9. Researcher: Let's listen to somebody else. Carlo 10. Carlo (he worked in a pair with Elsa): I wanted to say that on the axes it is written what they are, what you have to measure, look at, observe… 11. Researcher: Ok	The initiation of the discussion consists of focusing on the most chosen answer (remember here there is not only one correct answer). The teacher and the researcher choose to focus on answer F also because it is undoubtedly important and efficient to start the work on a graph with analysis of the axis Elsa and Carlo explain to their classmates that, by knowing which variables are represented on the axes, one can get much information on what is represented in the graph. Elsa and Carlo, therefore, turn themselves into resources for their classmates (FA strategy D)
12. Luca: Also on the axes… if it had been the contrary, with here (with gestures, he draws a vertical line) the time and here (with gestures, he draws a horizontal line) the distance, the graph would have changed… (he draws with gestures a possible new graph) 13. Researcher: Did you listen to what Luca said? (she is speaking to the other students) 14. Voices: Yes! 15. Researcher: I guess that somebody did not listen 16. Teacher: He said a very interesting thing 17. Researcher: Would you like to repeat what Luca said? (to Lavinia, who raised her hand) 18. Lavinia: We always have the distance from home (with gestures, she draws a vertical line) and time (with gestures, she draws a horizontal line), but maybe, in order to mislead us… 19. Teacher: Or because it is represented in another way. It could be! Then, it could be written… 20. Lavinia: In another way (with gestures, she draws the vertical and horizontal axes)	Luca points out that inverting the two variables represented on the axes leads to different graphs The researcher, in order to highlight Luca's intervention and to turn Luca into a real resource for the classmates (FA strategy D), carries out the following strategy: she asks if the other pupils listened to what Luca said, and asks another pupil to repeat it. This strategy makes Luca's thinking visible to the classmates Lavinia tries to repeat Luca's idea, activating herself as responsible for her learning (strategy E), but she speaks of "misleading" rather than of a different graph The teacher interrupts Lavinia and clarifies that there could be other possible graphs characterised by different variables represented on the two axes

(continued)

Table 3 (continued)

Excerpts from the class discussion	Analysis according to the FaSMEd three-dimensional framework
21. Researcher: Yes, that time is on the vertical axis and distance on the horizontal one. Luca said that, in that case, the graph changes. Or the same graph is interpreted in a different way. If I go to the blackboard… I was thinking of the impossible graph (in facing one of the preceding problem worksheets of the sequence, the class had worked on an impossible graph: time was represented on the horizontal axis, distance was represented on the vertical axis and the graph contained a vertical trait). Let's draw it… 22. Researcher: Ok. Now I draw only the vertical trait. If the time is here and the distance is here (she draws on the blackboard, see Fig. 4), we said that a vertical trait is impossible, isn't it? 23. Voices: Yes 24. Researcher: And if, instead, I put the time on the vertical axis, as Luca said, and the distance on the horizontal axis, would it be impossible? (she exchanges the variables on the axis in the same graph on the blackboard—see Fig. 5) 25. Voices: No! 26. Student: It is possible!	The researcher recalls to the students an impossible graph that was encountered in a previous problem worksheet. Her intervention is aimed at promoting a collective reflection on the fact that, if the variables change, the graph must be interpreted in another way We highlight the crucial choice of recalling a previous experience, occurred during the task sequence. In this way, the researcher and the teacher may also collect some feedback about the previous activities (strategy C)
27. Researcher: How would it be? Many students raise their hands 28. Researcher: Livio 29. Livio: In my opinion, it is impossible, because he does not move… because… if… (he points to the drawing on the blackboard) 30. Teacher: You can come to the blackboard Livio goes to the blackboard 31. Livio: If this is time (he points to the vertical axis), he spends this amount of time (he points to the vertical trait of the graph, Fig. 6), but he remains still, (he points to the horizontal axis, Fig. 7) in this trait of time (he points again to the vertical trait, Fig. 8)	At first, Livio asserts that the graph is impossible even when variables change; however, afterwards, he formulates a new interpretation Answering the teacher and going to the blackboard, expressing his ideas, Livio acts as responsible for his own learning (FA strategy E) and finally turns himself into a resource for his classmates (FA strategy D) We point out that Livio had not chosen option F when answering the poll. This means that the discussion enabled him to focus his attention on new specific aspects The new question posed by the researcher (concerning the new graph on the blackboard) makes Livio reflect on the importance of analysing the variables on the axes, and understand that the change of variables causes a change in interpretation of the graph

(continued)

Table 3 (continued)

Excerpts from the class discussion	Analysis according to the FaSMEd three-dimensional framework
32. Teacher: At this point … 33. Researcher: He remains still because the distance… 34. Livio: It is zero 35. Researcher: It is not zero 36. Livio: He is still here, he doesn't go there (with gestures he does a horizontal movement)	The aim of the teacher and the researcher is to make Livio realise why the graph on the blackboard represents that a person is still. In this way, they are activating him as a resource for his classmates (FA strategy D). In this phase of the discussion, Livio and the researcher are using the same word ("distance") with different meanings; in particular, Livio seems to interpret the word "distance" in terms of "walked distance" and not "distance from home"
37. Researcher: Let's listen to someone else 38. Teacher: Who agrees with him? Or on the contrary: who doesn't agree? 39. Researcher: Does somebody want to add something to what Livio said? Some students say no. Carlo raises his hand 40. Carlo: I wanted to say that he (Tommaso) spends some time still, at a given distance from home 41. Researcher: You wanted to add this. This idea is very interesting. We must not think that on the horizontal axis there is always the time and on the vertical axis the distance, we must be careful and check these aspects!	Carlo activates himself as a resource for Livio and his classmates (FA strategy D), proposing a correct interpretation of the word "distance" (in term of "distance from home") and clarifying the meaning of what Livio was trying to say (lines 34–36) Finally, the researcher gives feedback that moves the learners forward (FA strategy C), since she highlights the need to carefully analyse the variables represented on the axes as an efficient starting strategy for all activities involving interpretation of graphs

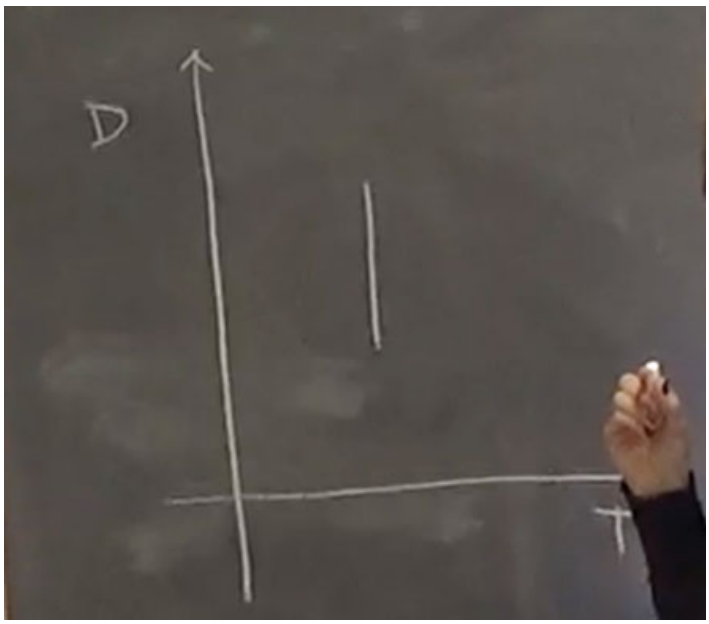


Fig. 4 The first graph drawn on the blackboard

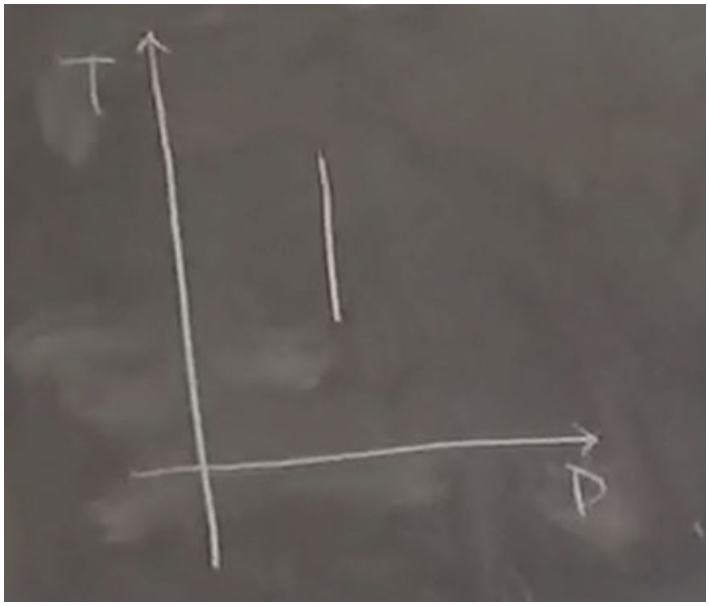


Fig. 5 The new graph on the blackboard

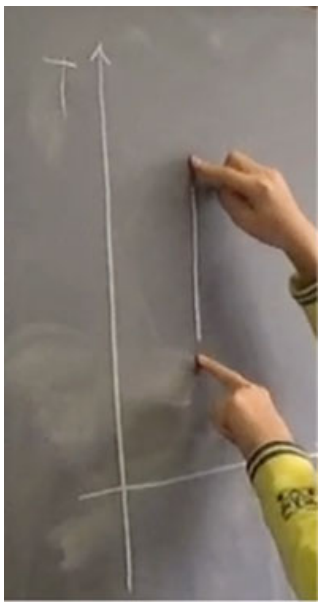


Fig. 6 Livio pointing to the vertical trait of the graph

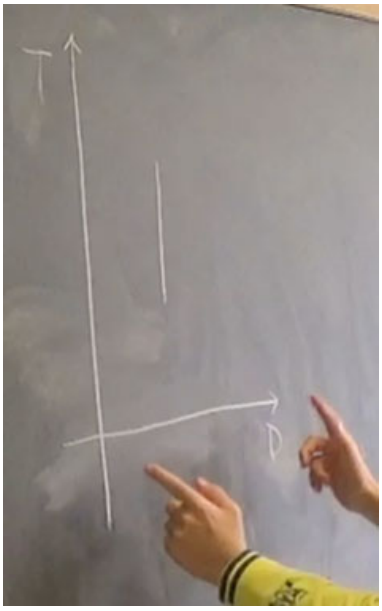


Fig. 7 Livio pointing to the horizontal axis

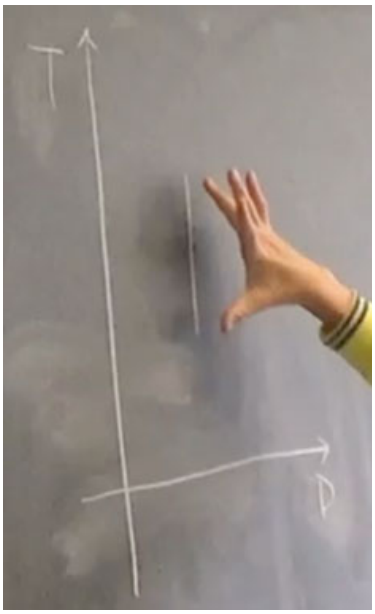


Fig. 8 Livio pointing again to the vertical trait

9 Discussion and Conclusions

Our work may be inserted into the stream of research concerning the use of technology, with a specific reference to Connected Classroom Technologies, for promoting FA. More specifically, in this chapter we focused on the use of polls to enhance effective classroom discussions with FA purposes at primary and lower secondary school levels. As outlined in our background paragraph, meta-reviews suggest the need for research on the use of CRS at lower school levels (Kay & Le Sage, 2009; Chien et al., 2016; Hunsu et al., 2016) and on the use of polls by groups of students (Fies & Marshall, 2006). Our research aimed at providing new insights in relation to these crucial issues.

We adopted a design-based approach and studied the use of polls for promoting FA in mathematical lessons through the "Processing and Analysing" and "Sending and Displaying" functionalities of the technology (Aldon et al., 2017). After presenting a classification of polls into four categories in relation to their content and their didactical aims, we illustrated two categories by means of the analysis of two episodes from our design experiments.

The episodes reported in examples 1 and 2 resonate with the main phases of the TEFA cycle proposed by Beatty and Gerace (2009): pose a challenging question; have students wrestle with the question and decide upon a response; use a CRS to collect responses and display a chart; elicit different reasons and justifications from students; develop a student-dominated discussion; provide meta-level comments. We remark a dialectic relationship between group work and poll: students work in small groups to establish a common answer to the poll, and students then discuss all together the results of the poll.

The analysis shows that the "Processing and Analysing" and "Sending and Displaying" functionalities of technology are efficient tools in enabling the teacher to share the results of polls with students and to structure around them a class discussion with FA purposes. This is in line with what research has identified as a crucial factor for promoting class discussion through the use of CRS: asking students to explain and justify their own answers (Chien et al., 2016). In these discussions, a complex variety of FA strategies emerged with the involvement of all the classroom actors (students, peers and teacher). Moreover, the overall data analysis (based on about 450 h of video-recording) allowed us to also highlight different structures of classroom discussions (and corresponding patterns of activated FA strategies) developed from polls. In the following we will present the main structures we identified, referring also to the two examples analysed in this paper.

First of all, we can distinguish between two main ways of initiating classroom discussions according to the percentages of students' answers in the case of polls with only one correct answer (categories 1 and 2).

When percentages of the correct answer and of one or more incorrect ones are quite balanced, the discussion asks students to compare these two (or more) options and to express the motivation for their choice. In this way, it is possible to focus on the mistakes that could lead to the choice of incorrect answers and to make students

activate themselves as owners of their own learning (FA strategy E). This is the case shown in example 1, which refers to a poll on argumentation in which the percentage for answer B (the correct one) is 50% and the percentage for answer C (incorrect) is 40%: as we can see from the excerpt, during the discussion the students could recognise their own mistakes and reflect on the reasons for them. In addition, students who chose the correct answer could benefit from the discussion, because they were asked to make their justification explicit; hence, they could develop their awareness about the reasons why they chose a specific option (again activation of FA strategy E). Furthermore, throughout the discussion, students are activated as instructional resources for their classmates (FA strategy D) because they give feedback to each other (FA strategy C) on the reasons why a chosen option is better than another.

When the percentages of students' answers are not balanced, the initiation of the discussion on the poll is different. After displaying the results of the poll, the teacher usually starts the discussion asking those who chose the incorrect answer to explain their choice (phase 1). This strategy is fruitful because it allows the students to focus on the mistakes that led to the choice of incorrect answers, making them activate themselves as owners of their own learning (FA strategy E). In this way, students can receive feedback about the task (FA strategy C) and are, thus, supported in recognising their own mistakes and reflecting on the reasons for them.

Afterwards, the teacher asks those who chose the correct answer (without revealing it is correct), to explain the reasons for their choice (phase 2). In this way, the students are led to focus on the justifications for the given answers and, then, can become more aware of the strategies they applied to solve the task (FA strategy E). As a consequence, the solving strategies are shared and discussed within the class. The students, activated as instructional resources for their classmates (FA strategy D), can then give feedback to each other (FA strategy C). Another important focus of this phase of the discussion is the comparison between the different ways of solving the task, in order to highlight the most efficient and to give feedback about the processing of the task. This kind of meta-level analysis is carried out to make students become aware of the most effective ways of facing specific tasks. In this way, feedback about self-regulation is provided and students could expand the "repertoire" of possible strategies to adopt when again facing similar problems.

During phases 1 and 2 of the discussion, some groups, who faced difficulties in choosing an answer, are asked to share their doubts and difficulties with their classmates. In this way, misconceptions are elicited and, consequently, students could give and receive feedback. Moreover, it often happens that some pairs/groups declare that they changed their mind during the discussion. During these phases of the discussion it is important to enable these students to share the reasons why they changed their mind, in order to activate themselves as owners of their own learning (FA strategy E).

The structure of classroom discussions is different when the discussions refer to a poll in which there is not one correct answer (categories 3 and 4).

Example 2, which refers to a poll on metacognitive aspects (in this case, efficient strategies to address problems containing graphs), illustrates a typical way of initiating the discussion in the case of polls belonging to categories 3 and 4. The discussion

is initiated with the analysis of the most chosen option in order to raise meta-level issues. The display of results makes some students revise their initial answer, thus making themselves responsible for their own learning (FA strategy E), or intervene to justify their choice, acting as resources for their classmates (FA strategy D). An example of this activation of FA strategies E and D is in lines 12–15, when Luca expresses his idea (FA strategy E) and the researcher relaunches it, so as to also involve the other students in the reflection and to turn Luca into a resource for his peers (FA strategy D). During the discussion, the teacher and the researcher intervene to give feedback that moves the students' learning forward (FA strategy C). This approach to the activation of classroom discussions is an instantiation of how CRS could be used in order to promote metacognitive strategies in classes, as advocated by research (Hunsu et al., 2016).

Another way of initiating (or developing) the discussion on metacognitive aspects [and, thus, promoting metacognitive strategies in the sense of Husu, Adesope, & Bayly, (2016)] is to focus on the options that where not chosen by students and to ask them to explain why they did not choose these. In this way, other meta-level issues could be faced because students are stimulated to reflect on the reasons why they preferred one option instead of another.

The episode presented in example 2 also represents an example of instant activation of a poll. Other frequent cases when the instant poll is initiated are situations where, during a classroom discussion on a problem worksheet, some incorrect answers emerge and the teacher decides to check whether all the students are aware of the non-correctness of these answers. This use of the poll may be linked to Beatty and Gerace's (2009) instructional purpose of eliciting a misconception. In the case in which this kind of poll is activated, students are asked to express their agreement with the incorrect solution, or to choose between different solutions (this kind of poll belongs to category 1 or 2); the results of the poll are displayed and subsequent discussion is aimed at working together towards a shared correct answer. Students who express the incorrect idea receive feedback (FA strategy C) and are led to become responsible for their own learning (FA strategy E); students who point out that the idea is not correct act also as resources for their classmates (FA strategy D).

Data analysis confirms that the teacher plays a crucial role in structuring the classroom discussion starting from polls to foster the activation of FA strategies. We believe that the different structures that we described, besides contributing to a theoretical reflection on the way polls may foster formative assessment, may also serve as a basis for guiding teacher practice and could be exploited in terms of teacher education. To this regard, we remark that in our design experiments the researcher collaborated with the teachers in developing effective classroom discussion. Further research is needed on the ways of supporting the teachers' autonomous use of this kind of resource.

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